

# Piter Z. Garcia Bautista

Rochester, NY | +1 (646) 288-3300 | pzg8794@rit.edu | pitergarcia@rochester.edu  
linkedin.com/in/garciapiterz | github.com/pzg8794 | garciapiterz.com

## Research Summary

---

NSF Robert Noyce Scholar and graduate researcher working at the intersection of equitable machine learning, reproducible experimentation, and high-stakes decision support. Current work spans fairness-aware learning, contextual and sequential decision workflows, and robust evaluation under shifting operational conditions. For maritime environmental monitoring, I am particularly interested in adaptive AI methods that remain reliable under noisy sensor streams, constrained communication, and non-stationary dynamics.

## Education

---

### Rochester Institute of Technology

Expected Aug 2026

*Master of Science, Data Science — GPA 3.9+*

*Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

### University of Rochester, Warner School of Education

Expected Aug 2026

*Master of Science track, Teaching Computer Science K-12*

*Rochester, NY*

- NSF Robert Noyce Teacher Scholarship Program (\$30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

### Rochester Institute of Technology

2015

*Master of Science, Computer Science*

*Rochester, NY*

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.

### Farmingdale State College

2012

*Bachelor of Science, Computer Engineering*

*Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

## Current Research Experience & Projects

---

### RIT & University of Rochester

2024 – Present

*Lead Researcher, EQUITAS Framework for Diagnostic Justice*

*Rochester, NY*

- Developed an interdisciplinary framework integrating fairness-aware ML, bioinformatics, and trauma-informed analysis.
- Built reproducible bias-assessment workflows and reported equity gaps across algorithmic variants.
- Coordinated cross-domain collaboration across computing, education, and health research contexts.

### Rochester Institute of Technology

2024 – Present

*Principal Investigator, Equitable Bioinformatics Project (ISTE-780)*

*Rochester, NY*

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines.

- Implemented reproducible evaluation codebase with automated metric reporting and statistical testing.

## **Rochester Institute of Technology**

*Graduate Assistant, Quantum and AI Research*

Fall 2025 – Present

*Rochester, NY*

- Run controlled experiments for contextual bandit and routing workflows under stochastic and adversarial settings.
- Build traceable experiment infrastructure across local, Colab, and GCP environments for robust comparison.

## **Teaching & Mentoring Experience**

---

### **Varsity Tutors & Independent Practice**

*STEM Tutor and Academic Mentor*

2021 – Present

*Remote & In-Person*

- Tutor Computer Science, Data Science, Mathematics, and Statistics for diverse learners.
- Design culturally responsive instruction and mentoring support for underrepresented students in STEM.

### **Rochester Institute of Technology**

*Faculty Assistant, Graduate Office*

2013 – 2015

*Rochester, NY*

- Supported graduate program operations, student documentation, and faculty coordination.

## **Selected Publications & Academic Contributions**

---

- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025).
- *Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework* (ED-447 Capstone, 2025).
- *RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements* (BIO-630 Final Project, 2025).
- *ASL Recognition Application: Image Processing for Sign Language Translation* (MS Thesis, 2015).

## **Technical Skills**

---

**Programming:** Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual learning workflows

**Data & Analysis:** pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking

**Infrastructure:** AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

## **Professional Experience**

---

### **VEDADATA Inc.**

*Data Solutions Engineer*

Sep 2022 – Aug 2024

*New York, NY*

- Designed and implemented data pipelines and analytics workflows using Python and cloud services.

**VIOME Inc.**

*AI Data Solutions Engineer*

Jan 2021 – Sep 2022

*New York, NY*

- Built ML workflows for healthcare applications with production deployment and monitoring support.

### **Selected Fit for NTNU AI-Enabled Maritime Systems Position**

---

- Strong fit for adaptive maritime monitoring research through experience with dynamic decision workflows, reliability analysis, and constrained-system evaluation.
- Demonstrated reproducibility discipline for multi-run experimental design, statistical validation, and transparent reporting.
- Interdisciplinary AI background spanning method development, applied data systems, and high-stakes deployment contexts.